YOLO11m 4-Class Textile Classifier Report

This document outlines the training and evaluation metrics for the updated YOLO11m classification model, trained on the 4-class textile dataset.

1. Model and Training Configuration

The model was trained utilizing transfer learning from the baseline weight `YOLO11m_benchmark/weights/best.pt` . The following configurations were applied during the process:

- **Model Architecture:** YOLO11m (Image Classification)
- **Task:** Classify
- **Total Classes:** 4
- **Input Image Size:** 640x640 pixels
- **Training Epochs:** 100
- **Batch Size:** 32
- **Optimizer:** Automatic selection
- **Patience:** 100 epochs (Early stopping was not triggered)
- **Data Augmentation:** Standard augmentations were disabled; utilized automatic augmentation (`randaugment`) with random erasing set to 0.4.

2. Training Performance Results

The training phase was executed for 100 epochs over the designated dataset. Convergence was properly established without critical signs of overfitting.

2.1 Key Metrics

- **Best Validation Accuracy (Top-1):** 0.93548 (recorded at Epochs 14 and 18)
- **Final Validation Accuracy (Top-1):** 0.91935 (at Epoch 100)
- **Final Training Loss:** 0.09135
- **Final Validation Loss:** 0.33362

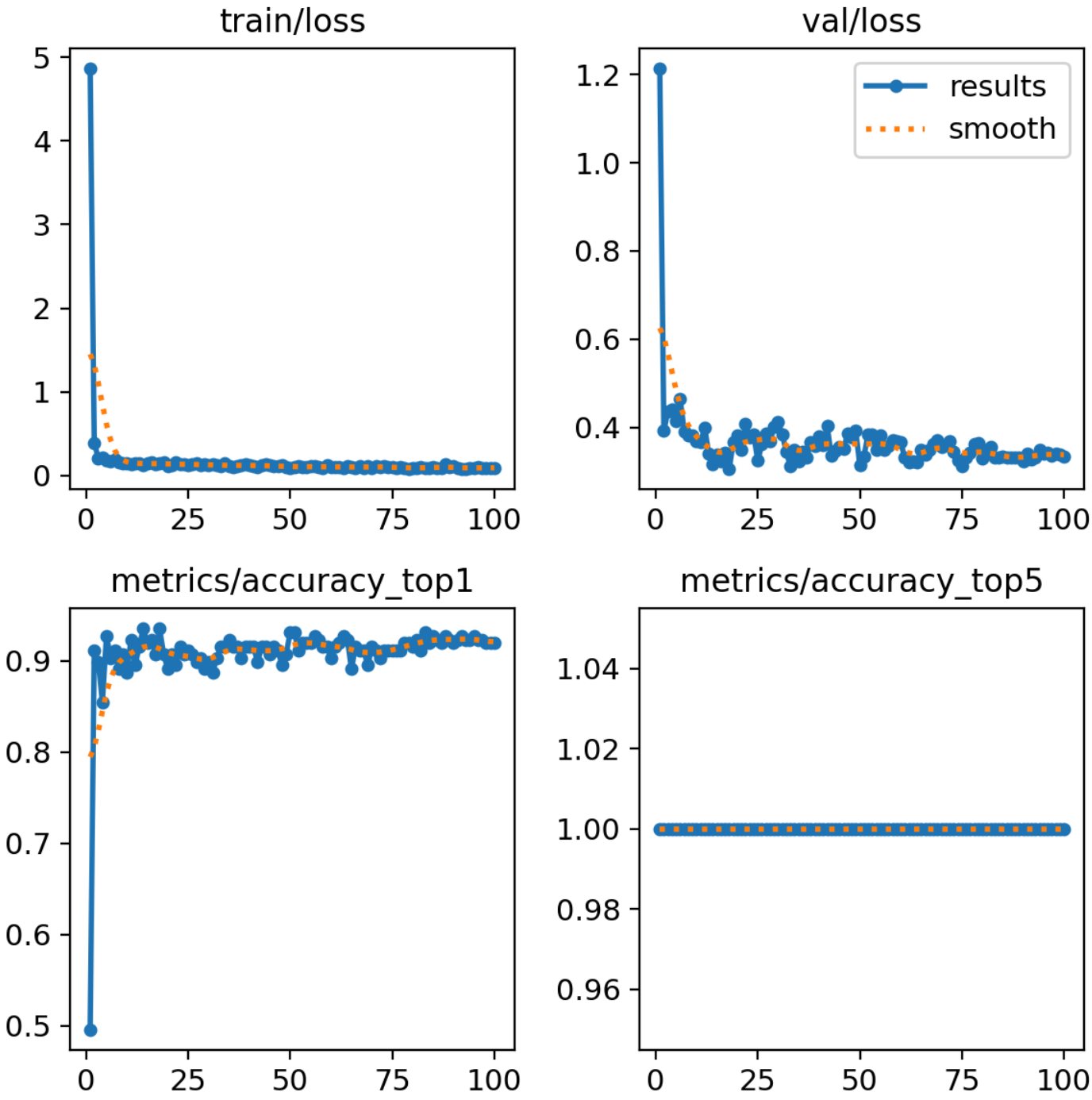
The loss trend demonstrated a rapid initial learning phase, stabilizing after the 10th epoch. The validation top-1 accuracy remained consistently in the ~91% to 93% range, indicating a robust internal representation of the 4 classes.

3. Visualizations

The following sections contain the epoch-by-epoch loss reduction, accuracy progressions, and confusion matrix data pinpointing classification distributions.

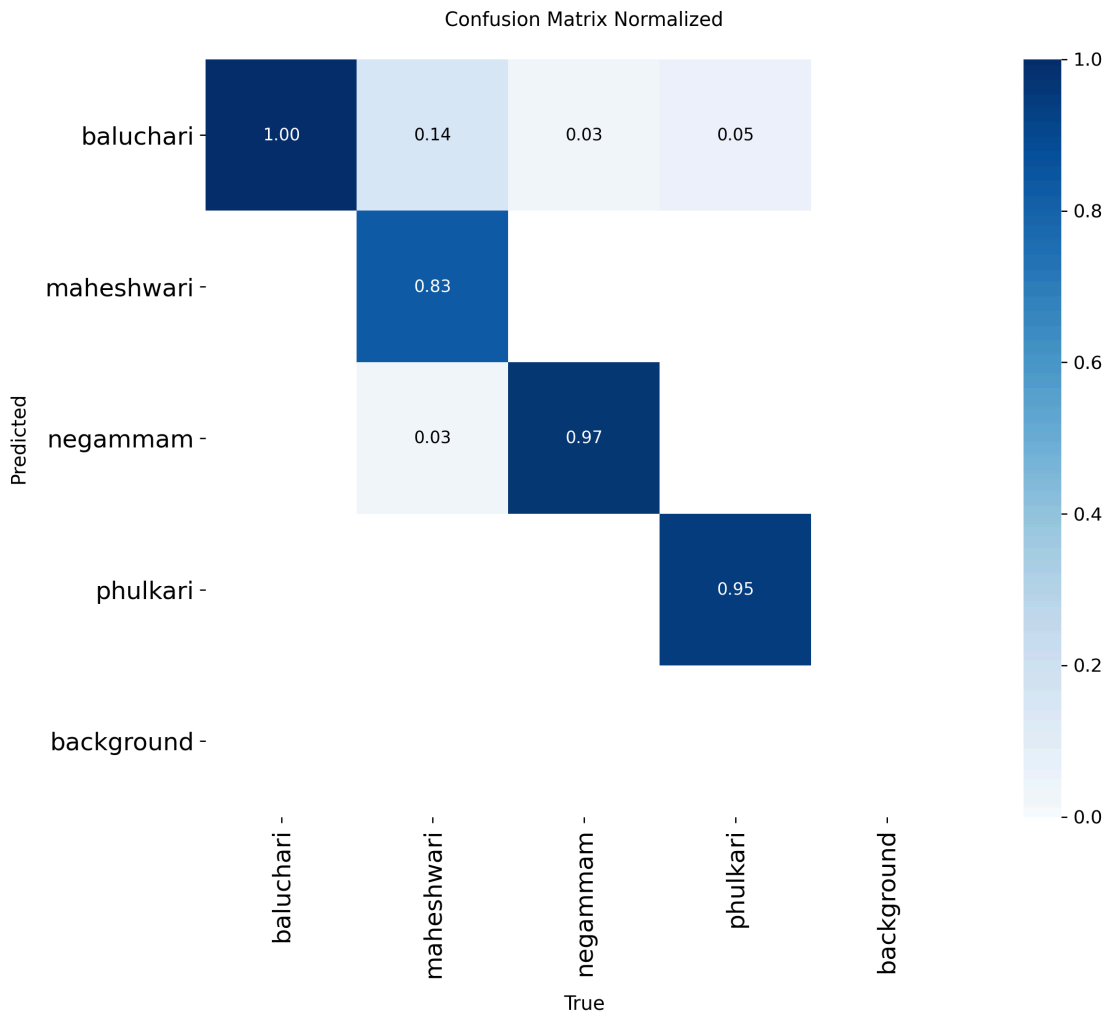
3.1 Epoch Performance Metrics

Below is the graphical representation of the training and validation loss, along with Top-1 and Top-5 accuracy progression over the 100 epochs.



3.2 Normalized Confusion Matrix

The normalized confusion matrix provides the proportion of accurate classifications versus misclassifications across the 4 textile classes.



3.3 Validation Predictions Sample

The image below displays a sample batch of validations containing the model predictions versus ground truth labels.

4. Conclusion

The transfer learning adaptation of the YOLO11m model for the 4-class textile dataset achieved strong performance. The peak top-1 accuracy reached approximately 93.5%. With stable validation loss behavior during the final training stages, the resulting weights located at `weights/best.pt` provide a highly reliable baseline for the classification application.